

DAY 2 — COMPLETE LEARNING LOG

Networking, Linux, Remote Connectivity & Cloud Fundamentals

1. Subnetting

Subnetting is the process of dividing a larger network into smaller logical networks called subnets. It helps organize networks, manage IP addresses and control network traffic.

2. Supernetting

Supernetting is the process of combining multiple smaller networks into a larger network or address block. It is commonly associated with route aggregation and helps reduce the number of routing table entries.

3. Routing

Routing is the systematic process of selecting paths for data packets to travel from a source network to a destination network. Routers use routing information to forward packets.

4. Protocols & Routing Protocols

A protocol is a set of rules that defines how devices communicate. Routing protocols allow routers to exchange routing information and determine suitable paths.

Protocol	Full name / description
RIPv1	Routing Information Protocol, version 1
RIPv2	Routing Information Protocol, version 2
EIGRP	Enhanced Interior Gateway Routing Protocol
IGRP	Interior Gateway Routing Protocol
BGP	Border Gateway Protocol

5. Port Numbers

A port number is a logical number used to identify a specific application or service on a computer. Port numbers work with transport protocols such as TCP and UDP.

Range	Name	Purpose
0–1023	Well-known ports	Common standard services
1024–49151	Registered ports	Registered applications/services
49152–65535	Dynamic / private ports	Temporary client-side or application communication

6. Private IP vs Public IP

Private IP addresses are used inside local/private networks and are not directly routable on the public Internet. Public IP addresses are globally routable and can be used for Internet communication.

Private IP	Public IP
Used within private/local networks	Used on public networks/Internet
Not directly routable over the Internet	Routable over the Internet
Usually assigned by a local network administrator or DHCP	Assigned by an Internet/service provider or organization
Can be reused in different private networks	Must be globally unique when used on the Internet
Common private ranges: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/24	Public addresses are allocated from globally managed address space

7. Private IP Address Ranges

- **Class A:** 10.0.0.0 – 10.255.255.255
- **Class B:** 172.16.0.0 – 172.31.255.255
- **Class C:** 192.168.0.0 – 192.168.255.255

8. Subnet Mask

A subnet mask is used with an IP address to identify the network portion and host portion of an IPv4 address. It helps determine which devices belong to the same subnet.

9. How Does Internet Traffic Travel?

When a device sends data to an Internet destination, the data is divided into packets. The packets are forwarded through local networking equipment and routers, with routing decisions made at each network hop until they reach the destination.

10. Linux Basics

Linux is an open-source operating system family widely used in servers, cloud infrastructure, development environments and many other systems.

11. Linux Shell

A shell is a command interpreter that provides an interface between the user and the operating system. It accepts commands and executes them.

Shell	Full name / common use
Bash	Bourne Again Shell — general-purpose shell and scripting
sh	Bourne Shell — basic shell scripting
zsh	Z Shell — interactive use and customization
Fish	Friendly Interactive Shell — user-friendly interactive shell
ksh	KornShell — scripting and interactive use
Tcsh	TENEX C Shell — C-style shell interaction
Dash	Debian Almquist Shell — fast system shell

12. Linux Commands & Connectivity

- **SSH:** Secure Shell, used for secure remote login and command execution.
- **SSH default port:** TCP 22.
- **PuTTY:** a commonly used SSH/Telnet client, especially on Windows.
- **RDP:** Remote Desktop Protocol, commonly used for remote Windows desktop access.
- **RDP default port:** TCP 3389.
- **PowerShell:** a Windows command-line shell and scripting environment.
- **Bash:** commonly used on Linux systems for command execution and scripting.

13. Remote Connectivity Examples

Windows → Linux: commonly connect using an SSH client such as PuTTY or another SSH client, using TCP port 22.

Windows → Windows: Remote Desktop Protocol (RDP) can be used for remote desktop access, commonly through TCP port 3389.

Mac → Linux: SSH can be used from the terminal to connect securely to a Linux system.

Linux → Linux: SSH is commonly used for secure remote administration.

14. Cloud Computing

Cloud computing is the delivery of computing resources and services over a network, typically the Internet. Cloud platforms provide resources such as compute, storage, networking, databases, security and identity management.

- **AWS:** Amazon Web Services.
- **Microsoft Azure:** Microsoft's cloud computing platform.
- **Google Cloud:** Google's cloud computing platform.
- Common cloud resources include compute, storage, databases, virtual networks, containers, security and identity/access management.

15. Cloud Regions & Availability Zones

A cloud region is a geographic area containing cloud infrastructure. Within a region, availability zones are separate infrastructure locations designed to provide isolation and resilience.

16. Cloud Cost Concepts

- Cloud services commonly use usage-based or pay-as-you-go pricing.
- Costs can depend on compute usage, storage, data transfer and other resources.
- Cloud pricing and resource availability can vary by region and service.

17. Day 2 Learning Outcome

By the end of Day 2, the learning covered subnetting, supernetting, routing, routing protocols, port numbers, private and public IP addresses, subnet masks, Internet traffic flow, Linux and shells, SSH/PuTTY, Windows remote connectivity, and cloud computing concepts including AWS, Azure, Google Cloud, regions and availability zones.

Suggested GitHub commit: docs: add Day 2 networking, Linux and cloud learning log

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